

HEALTHCONNECT EXPERIENCE LAB

WEEK 5 — PROJECT SUMMARY

Data Analytics Track

1. Planned Work

At the beginning of Week 5, the main objective was to move from the exploratory planning conducted during Week 4 to the practical implementation of the Data Analytics workstream.

The planned activities were:

- prepare and validate the HealthConnect appointment dataset;
- inspect data types, missing values, duplicates and inconsistencies;
- perform exploratory data analysis;
- calculate and validate key performance indicators;
- identify the main factors associated with missed appointments;
- develop a Power BI analytical dashboard;
- generate actionable business insights;
- formulate recommendations for HealthConnect;
- document limitations and define the next steps for Week 6.

This follows the official Week 5 transition from planning to analysis, development and initial implementation.

2. Work Actually Completed

During Week 5, the planned Data Analytics activities were implemented using the **HealthConnect Appointment Data**.

Data preparation

The dataset was inspected and validated.

The analysis confirmed:

- **5,000 appointment records;**
- unique appointment identifiers;
- no duplicate appointment records;
- coherent booking and appointment dates;
- consistency between **booking_lead_days** and the difference between booking and appointment dates;
- three variables containing missing values.

Data transformation

Power Query was used to:

- correct date data types;
- handle missing values;
- create meaningful categories for booking lead time;
- create distance categories;
- prepare the dataset for analysis in Power BI.

The original dataset was preserved, while transformed data and derived variables were handled separately, in accordance with the Week 5 requirement not to overwrite the original project resources.

3. Exploratory Data Analysis

The EDA focused on identifying variables associated with appointment No-Shows.

The following dimensions were analysed:

- booking lead time;
- previous No-Shows;
- distance to the clinic;
- reminder status and channel;
- appointment type;
- age group;
- gender.

The analysis showed that the strongest observed differences were associated with:

1. **booking lead time;**
2. **previous No-Shows;**
3. **distance to the clinic.**

The reminder variable was also examined as a potentially relevant operational factor.

4. KPI Development

Five indicators were developed to monitor appointment attendance:

KPI	Result
Total appointments	5,000
No-Shows	2,423
No-Show rate	48.46 %
Attended appointments	2,314
No-Show rate excluding cancellations	51.16 %

The main KPI retained for the dashboard is the **48.46 % overall No-Show rate**, ensuring consistency with the initial analysis.

The official Week 5 requirements specifically ask the Data Analytics track to develop and interpret 3–5 KPIs.

5. Dashboard Development

A Power BI dashboard was developed to provide an analytical view of appointment No-Shows.

The dashboard includes:

- headline KPIs;
- No-Show rate according to booking lead time;
- No-Show rate according to previous No-Shows;
- No-Show rate according to distance;
- No-Show rate according to reminder channel;
- complementary demographic and appointment analyses.

The dashboard was subsequently reviewed to improve:

- visual hierarchy;
- chart ordering;
- labels;
- business-oriented titles;
- interpretation of the results;
- emphasis on the most important analytical findings.

6. Main Results

Five major insights were identified.

Insight 1 — Booking lead time

The No-Show rate increases substantially as the booking lead time increases, from **27.81 % for 0–7 days to 60.49 % for appointments booked more than 30 days in advance.**

Insight 2 — Previous No-Shows

Patients with previous No-Shows have progressively higher observed No-Show rates, reaching **68.82 % among patients with 3+ previous No-Shows.**

Insight 3 — Distance

Distance shows a relatively limited difference below 20 km, while the observed No-Show rate increases to **57.76 % for patients located more than 20 km from the clinic.**

Insight 4 — Reminder channels

The observed No-Show rate is **47.36 % among patients who received a reminder compared with 51.39 % among those who did not.** SMS has the lowest observed No-Show rate among the reminder channels analysed.

However, these results represent **associations rather than proof of a causal effect**.

Insight 5 — Combined factors

The cross-analysis indicates that the highest observed No-Show rates occur when a long booking lead time is combined with previous No-Show history.

This suggests that HealthConnect could consider a combined prioritisation approach rather than evaluating each factor independently.

7. Recommendations

Based on the findings, the following actions are recommended:

1. Strengthen follow-up for appointments booked more than 30 days in advance.

These appointments show a substantially higher No-Show rate.

2. Prioritise patients with repeated previous No-Shows.

A history of previous No-Shows is strongly associated with higher observed No-Show rates.

3. Pay particular attention to patients located more than 20 km from the clinic.

Additional confirmation or accessibility-oriented measures could be considered for this group.

4. Evaluate reminder strategies.

SMS showed the lowest observed No-Show rate among the analysed reminder channels, but controlled testing would be necessary before attributing an actual causal effect.

5. Develop a combined prioritisation approach.

HealthConnect could combine:

booking lead time + previous No-Shows + distance

to identify appointments requiring greater follow-up attention.

8. Challenges Encountered

Several challenges were encountered during implementation.

Date formatting

The booking and appointment dates were initially interpreted as text because of their date format. This required a correction of the data type and regional settings in Power Query.

Missing values

Missing values were identified in `reminder_channel`, `distance_to_clinic_km` and `waiting_time_minutes`.

Different treatments were required depending on the meaning of the variable.

Interpretation of reminder data

The reminder analysis initially raised the question of whether reminders directly reduce No-Shows.

The analysis was therefore refined to distinguish **association from causality**, avoiding a causal conclusion that the available data cannot support.

Dashboard design

The first dashboard version contained the necessary information but required improvement in visual hierarchy, readability and business storytelling.

The dashboard was subsequently redesigned around the most important analytical findings.

9. Important Analytical Decisions

Several decisions were made during implementation.

Decision 1 — Power BI as the main visualization tool

Power BI was retained as the primary tool for KPI calculation and dashboard development because it allows the results to be communicated interactively to business users.

Decision 2 — Creation of analytical categories

Continuous variables such as booking lead time and distance were grouped into meaningful categories to facilitate interpretation.

Decision 3 — Median imputation

Median values were used for the limited missing observations in distance and waiting time.

Decision 4 — "Aucun" for missing reminder channel

Missing `reminder_channel` values correspond to appointments where no reminder was sent. They were therefore represented as **"Aucun"** rather than treated as an unknown category.

Decision 5 — Descriptive interpretation

The analysis was deliberately kept descriptive. No causal claim or individual-level prediction was made from the observed relationships.

10. Changes from Week 4

The Week 4 work established the initial hypotheses and identified potentially important variables.

During Week 5, these hypotheses were tested more systematically through EDA.

The analysis confirmed that **booking lead time** and **previous No-Show history** were particularly important dimensions.

The reminder analysis was also refined: rather than describing reminder channels as having a proven "impact", the results are now presented as observed associations.

The dashboard was also evolved from a collection of analytical visuals toward a more business-oriented decision-support tool.

11. Collaboration and Dependencies

The Data Analytics workstream has an important potential dependency with the **Data Science track**.

The EDA identified several variables that could be relevant for a future predictive analysis, particularly:

- booking lead time;
- previous No-Shows;
- distance to the clinic;
- reminder information.

These findings can provide useful input for future feature selection or modelling.

Important: this section should only claim an actual collaboration if you genuinely exchanged with a Data Science teammate. Otherwise, it should be described as an **identified dependency/opportunity for coordination**, not as completed collaboration.

The Week 5 requirements explicitly include collaboration or coordination with other project tracks where relevant.

12. Risks and Limitations

The main limitations identified are:

- the dataset is fictional and anonymised;
- the analysis is primarily descriptive;
- observed associations do not establish causality;
- some categories contain relatively small numbers of observations;
- imputation may slightly influence some results;
- reminder groups may differ in ways that are not controlled for;
- no predictive model has yet been developed.

These limitations should remain visible when communicating the findings.

13. Work Remaining

The following activities remain for the next phase:

1. finalise the Power BI dashboard;
2. validate all dashboard values against the cleaned dataset;
3. finalise the analytical documentation;
4. organise supporting project files;
5. continue coordination with other project tracks;
6. investigate whether the identified factors can support a more advanced analysis;
7. prepare the next phase of the HealthConnect solution.

14. Objective for Week 6

The main objective for Week 6 should be to **build on the validated Week 5 findings and move toward a more advanced decision-support approach.**

For the Data Analytics track, this can include:

Using the factors identified during Week 5 to deepen the analysis, validate the most important patterns and support the development of more targeted strategies for reducing No-Shows.

If the broader project requires a predictive component in Week 6, the variables identified during this week's EDA can serve as a starting point for coordination with the Data Science track.